

Assignment A3

Analyzing Helmholtz Resonances

SCIE 001 Physics

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Introduction

If you blow across the opening of an empty bottle, you can hear a distinctly loud and low pitch sound. This is an example of a Helmholtz resonance, which is due to air in and near the neck of the bottle vibrating. You can model this vibration as a chunk of air in the neck of the bottle being pushed and pulled by the rest of the air below it (which is compressing and expanding like a spring). This phenomenon can therefore be simply understood in terms of simple harmonic motion.

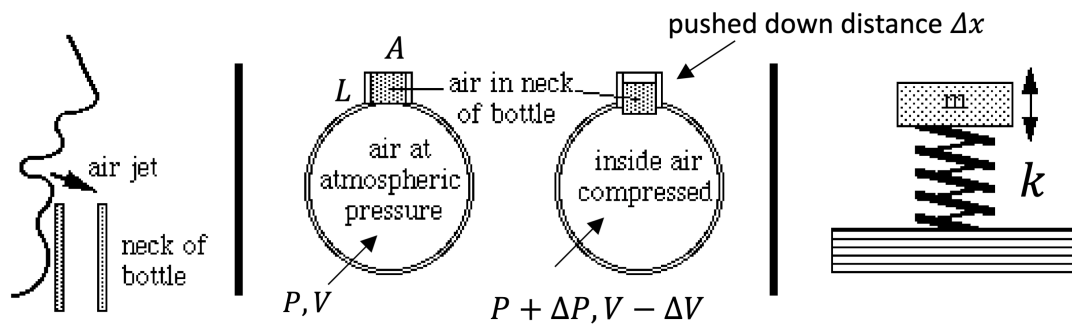


Figure 1: (left) Blowing across the opening of an empty bottle makes the air in and near the neck of the bottle oscillate, which you can hear as a loud low pitch sound. (middle and right) You can model this effect by treating the air in the neck of the bottle as a cylindrical mass m of cross-sectional area A and length L that is oscillating above an ideal spring with spring constant k . **Adapted from:** <https://newt.phys.unsw.edu.au/jw/Helmholtz.html> (check it out).

This assignment has two parts: in the first, you will derive a simple theoretical model for how a bottle's Helmholtz resonance changes with the volume of air in the bottle. In the second part, you will collect your own data and plot your theoretical model with your data to assess its validity.

Part 1 - The Model

1. [1 pt] Imagine the air in the neck of the bottle is a small cylindrical mass m oscillating due to an ideal spring with spring constant k (as shown in Figure 1). When the mass is pushed down and then released, it starts oscillating (and creates the sound that we hear). What is the frequency f of this oscillation in terms of the air density ρ_{air} , A , L and k ?

In an ideal spring system with spring constant k and mass m , we know:

$$\omega = \sqrt{\frac{k}{m}} = 2\pi f$$

Thinking about the bottle as an ideal spring system, we can form a relationship between the air in the neck and the mass on the spring :

$$m = LA\rho_{\text{air}}$$

So:

$$\sqrt{\frac{k}{LA\rho_{\text{air}}}} = 2\pi f$$

$$f = \frac{\sqrt{\frac{k}{LA\rho_{\text{air}}}}}{2\pi}$$

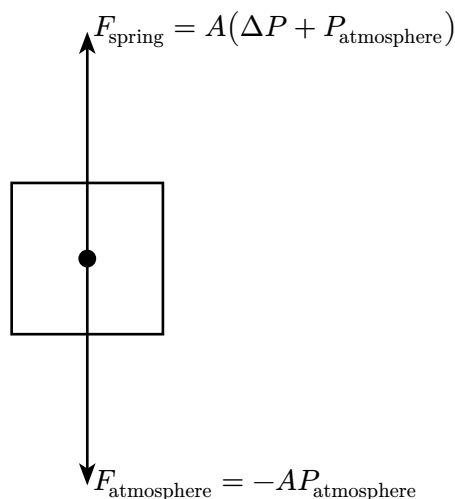
2. [2 pt] We know the spring models the volume of air below the neck of the bottle compressing and expanding. The goal is now to relate k to the parameters of this volume of air (as shown in Figure 1). We can do this by using Hooke's law, which states:

$$k = \left| \frac{F}{\Delta x} \right|$$

- a) When the mass m is pushed down by some small Δx , the air pressure in the bottle below it increases by some small amount ΔP . Draw a FBD clearly indicating the vertical forces acting on this mass of air in the neck of the bottle. (Hint: Your forces should only depend on atmospheric pressure, ΔP and A .) What is the net force F on this air mass? Be clear on the direction of this net force.

Assuming the internal pressure of the bottle is initially at equilibrium with atmospheric pressure.

Where up is positive:



$$\begin{aligned}
 F_{\text{net}} &= F_{\text{spring}} + F_{\text{atmosphere}} \\
 &= A(\Delta P + P_{\text{atmosphere}}) - AP_{\text{atmosphere}} \\
 F_{\text{net}} &= A\Delta P + AP_{\text{atmosphere}} - AP_{\text{atmosphere}}
 \end{aligned}$$

$$\therefore F_{\text{net}} = A\Delta P$$

- b) When the mass is pushed down by some small Δx , the volume of air in the bottle below it decreases by some small amount ΔV . Express this small volume change in terms of Δx and A .

$$\Delta V = A\Delta x$$

- c) Show that in the limit of infinitesimal changes in pressure and volume, the spring constant is given by $k = \frac{dP}{dV} A^2$. Show that the units make sense.

$$\Delta V = A \Delta x$$

$$\Delta x = \frac{\Delta V}{A}$$

$$k = \left| \frac{F}{\Delta x} \right|$$

$$k = \left| \frac{A \Delta P}{\frac{\Delta V}{A}} \right|$$

$$k = \left| \frac{A^2 \Delta P}{\Delta V} \right|$$

Noting:

$$\Delta P = P(V + \Delta V) - P(V)$$

Where $P(V)$ is the pressure of the air in the bottle at volume V .

Taking the limit of our expression for k as ΔV approaches 0:

$$\begin{aligned} \lim_{\Delta V \rightarrow 0} k &= \lim_{\Delta V \rightarrow 0} \left| \frac{A^2 (P(V + \Delta V) - P(V))}{\Delta V} \right| \\ k &= A^2 \left| \lim_{\Delta V \rightarrow 0} \frac{(P(V + \Delta V) - P(V))}{\Delta V} \right| \quad A^2 \geq 0 \end{aligned}$$

Rewriting in derivative notation:

$$\boxed{\therefore k = A^2 \left| \frac{dP}{dV} \right|}$$

Showing that the units make sense:

Left hand:

$$k = \frac{[N]}{[m]}$$

Right hand:

$$A^2 \left| \frac{dP}{dV} \right| = [m^2]^2 \frac{\frac{[N]}{[m]^2}}{[m^3]} = \frac{[N][m]^4}{[m]^5} = \frac{[N]}{[m]}$$

Left hand equals right hand, therefore the units make sense.

3. [2 pts] We now need to determine how pressure changes with volume in this system to solve the derivative in the expression for k . Luckily this is something that we investigated thoroughly last semester.

- a) In 2-3 sentences, explain in your own words why an isothermal relationship (i.e. PV constant) is not appropriate for this system while an adiabatic relationship is.

An isothermal relationship assumes that the temperature of the air on the inside of the bottle stays constant. This is not true for our system - since as we are compressing the air in the bottle relatively quickly, its temperature rises and does not have enough time to return to thermal equilibrium with the environment (dissipate its heat). A better approximation would be that the air loses no heat to the environment in the time that we compress it - an adiabatic process.

- b) For adiabatic processes, PV^γ constant. Differentiate this equation to find an equation for $\frac{dP}{dV}$.

Differentiating:

$$\begin{aligned}
 PV^\gamma &= \text{constant} \\
 \frac{d}{dV}(PV^\gamma) &= 0 \\
 \left(\frac{d}{dV}P\right)V^\gamma + P\left(\frac{d}{dV}V^\gamma\right) &= 0 \\
 \left(\frac{dP}{dV}\right)V^\gamma + P\gamma V^{\gamma-1} &= 0 \\
 \frac{dP}{dV} &= \frac{-P\gamma V^{\gamma-1}}{V^\gamma} = -P\gamma V^{-1} \\
 \boxed{\therefore \frac{dP}{dV} &= \frac{-P\gamma}{V}}
 \end{aligned}$$

4. [0.5 pts] Using your answers for the previous questions, write an equation for the frequency f of our Helmholtz oscillator in terms of γ , ρ , A , L , P and V .

$$\begin{aligned}
 k &= A^2 \left| \frac{dP}{dV} \right| \quad \frac{dP}{dV} = \frac{-P\gamma}{V} \\
 k &= A^2 \left| \frac{-P\gamma}{V} \right| \\
 k &= \frac{A^2 P \gamma}{V} \quad \frac{P\gamma}{V} > 0 \\
 f &= \frac{\sqrt{\frac{k}{LA\rho_{\text{air}}}}}{2\pi} \quad \text{from question 1} \\
 f &= \frac{\sqrt{\frac{\left(\frac{A^2 P \gamma}{V}\right)}{LA\rho_{\text{air}}}}}{2\pi} \\
 \boxed{f &= \frac{\sqrt{\frac{AP\gamma}{VL\rho_{\text{air}}}}}{2\pi}}
 \end{aligned}$$

5. [0.5 pts] You can simplify your result by recognizing that the speed of sound in air v is given by:

$$v = \sqrt{\gamma \frac{P}{\rho_{\text{air}}}}$$

Rewrite your equation in terms of v , A , L , and V . Show that the units make sense.

$$f = \frac{\sqrt{\frac{AP\gamma}{VL\rho_{\text{air}}}}}{2\pi} = \frac{\sqrt{\left(\frac{A}{VL}\right)\left(\gamma \frac{P}{\rho_{\text{air}}}\right)}}{2\pi} \quad v^2 = \gamma \frac{P}{\rho_{\text{air}}}$$

$$f = \frac{\sqrt{\left(\frac{A}{VL}\right)(v^2)}}{2\pi}$$

$$\therefore f = \frac{\sqrt{\frac{Av^2}{VL}}}{2\pi}$$

Showing that the units make sense:

Left hand:

$$f = [\text{Hz}] = [\text{s}^{-1}]$$

Right hand:

$$\frac{\sqrt{\frac{Av^2}{VL}}}{2\pi} = \sqrt{\frac{[\text{m}^2][\text{m/s}]^2}{[\text{m}^3][\text{m}]}} = \sqrt{\frac{\left[\frac{\text{m}^4}{\text{s}^2}\right]}{[\text{m}^4]}} = \sqrt{\frac{1}{[\text{s}^2]}} = \frac{1}{[\text{s}]} = [\text{s}^{-1}]$$

The units of both sides are the same, therefore the units make sense.

6. [4 pts] In preparation for the second part of this assignment, explain at least two modeling approximations or assumptions (besides the adiabatic condition) that you made above and why they may or may not be reasonable for your experiment. Take up to 100 words max and provide citations if needed.

We assume that the air acting as the mass fills the entire neck cavity, and that the air acting as a spring fills the entire body. It may be that the division between oscillating and springing lies somewhere other than the neckline, or is less well defined altogether (air being both pushed down, and resisting compression simultaneously).

We also assume that this division does not change with the resonance frequency or the volume remaining. This is not reasonable, as once the water in the bottle reaches the neck, our model breaks down, but we can still hear sound.

Words: 98

Part 2 - The Experiment

[10 pts - see rubric for further details]

1. [2 pt] Find a glass bottle with a well-defined neck such as the wine or ginger beer bottles, illustrated in the image below. You need to measure the volume below the neck, the cross-sectional area A of the neck and its length L .

Fill out this table of important parameters for your bottle and include a picture of your bottle below with clear annotations of relevant dimensions.

Nominal volume of bottle	0.773100 ± 0.000041 L
Estimated volume of air below the neck of bottle (when empty)	0.741699 ± 0.000041 L
Length of neck	8.520 ± 0.11 cm
Diameter of neck	1.85 ± 0.050 cm
Local speed of sound used in model	343.21 m/s [1]
Density of water used	1.00 kg / L [2]

Table 1: Bottle measurements and model parameters.

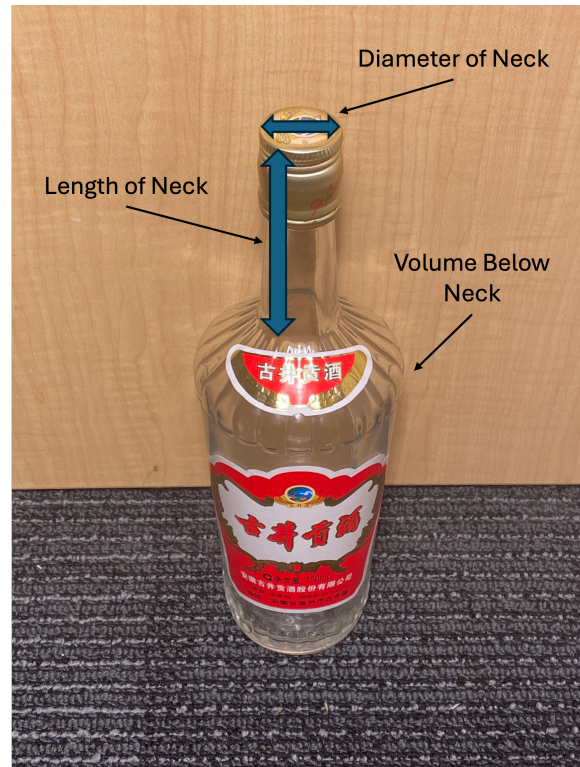


Figure 2: An image of the bottle annotated with relevant dimensions.

For uncertainty in the length and diameter, gaussian uncertainty was used. For gaussian uncertainty we use a 95% confidence interval divided by 4.

$$u[x] = \frac{x_{\max} - x_{\min}}{4}$$

Since the bottle's volume was determined by water weight (as described in 3a), the uncertainties needed to be propagated:

The scale used rounded to the nearest 0.1 g, so we can use rectangular uncertainty.

$$u[m] = \frac{\text{"width" of rounding}}{2\sqrt{3}} = \frac{0.1 \text{ g}}{2\sqrt{3}} = 0.029 \text{ g}$$

Now we subtract the empty bottle weight, from the full bottle weight, so:

$$m_{\text{water}} = m_{\text{empty}} - m_{\text{full}}$$

$$u[m_{\text{water}}] = \sqrt{\left(u[m_{\text{empty}}] \frac{\partial}{\partial m_{\text{empty}}} (m_{\text{empty}} - m_{\text{full}})\right)^2 + \left(u[m_{\text{full}}] \frac{\partial}{\partial m_{\text{full}}} (m_{\text{empty}} - m_{\text{full}})\right)^2}$$

$$\begin{aligned} u[m_{\text{water}}] &= \sqrt{u[m_{\text{empty}}]^2 + u[m_{\text{full}}]^2} \\ &= \sqrt{(0.029 \text{ g})^2 + (0.029 \text{ g})^2} \end{aligned}$$

$$u[m_{\text{water}}] = 0.041 \text{ g}$$

Now to turn it into volume, we divide by the density of water ρ_{water} .

$$\begin{aligned} V &= \frac{m}{\rho_{\text{water}}} \\ u[V] &= \frac{u[m]}{\rho_{\text{water}}} \\ u[V] &= \frac{0.041 \text{ g}}{\left(\frac{1000 \text{ g}}{1 \text{ L}}\right)} = 0.000041 \text{ L} \end{aligned}$$

This is the same uncertainty for estimating the volume of air below the neck of the bottle.

2. [2 pts] Blow over the top of your bottle and obtain the frequency spectrum of the sound emitted using the Audio Spectrum function of the PhyPhox app.

- a) Include a screenshot of your measurement in the "Spectrum" tab below. At what frequency f is the Helmholtz resonance for your empty bottle? Explain in 2-3 sentences how you can tell.

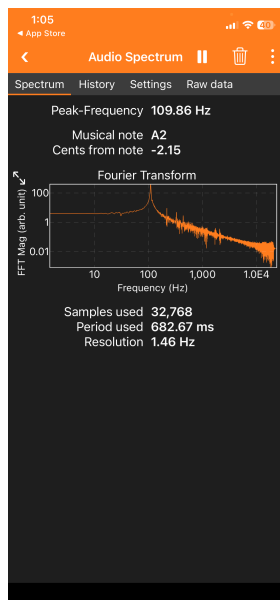


Figure 3: Screenshot of PhyPhox while blowing into the empty bottle.

The Helmholtz resonance frequency for my empty bottle is $109.89 \pm 0.42 \text{ Hz}$.

You can tell that it is 109.89 Hz because the fourier transform of the sound from my bottle shows a distinct peak at that frequency (also noted in the Peak-Frequency number above the graph).

The uncertainty comes from the resolution of 1.46 Hz shown below the graph. Since this is a discrete digital measurement we can use rectangular uncertainty:

$$u[f] = \frac{\text{"width" of rounding}}{2\sqrt{3}} = \frac{1.46 \text{ Hz}}{2\sqrt{3}} = 0.42 \text{ Hz}$$

- b) In 2-3 sentences, explain the trade-off between the frequency resolution of the obtained spectrum and the number of samples taken. Hint: you can change the number of samples in the “Settings” tab. What settings did you use for your experiment?

The more samples that you take for each fourier transform, the more frequency resolution you have. This makes intuitive sense, as with more samples you give more information to the fourier transform, allowing it to distinguish better between similar frequencies.

A lower sampling rate would provide less frequency resolution than a higher sampling rate, so I used the highest possible sampling rate of 32768 samples over a period of 682.67 ms for this experiment.

3. [4 pts] Now add some water to your bottle and measure the frequency again for various amounts of water in the bottle.

- a) Make a plot of the resonant frequency f against the remaining volume of air V in the bottle. Briefly explain how you measured and calculated V below.

Volume was calculated using water-weight. First, the bottle was weighted empty, then weighed with water filled to the base of the neck. Subtracting these values and dividing by the density of water yielded the volume of the body of the bottle.

Then for each trial, water was added to the bottle and the apparatus was weighed. Subtracting the weight of the empty bottle and dividing by the density of water, then subtracting that from the volume of the body yielded the volume of air under the neck.

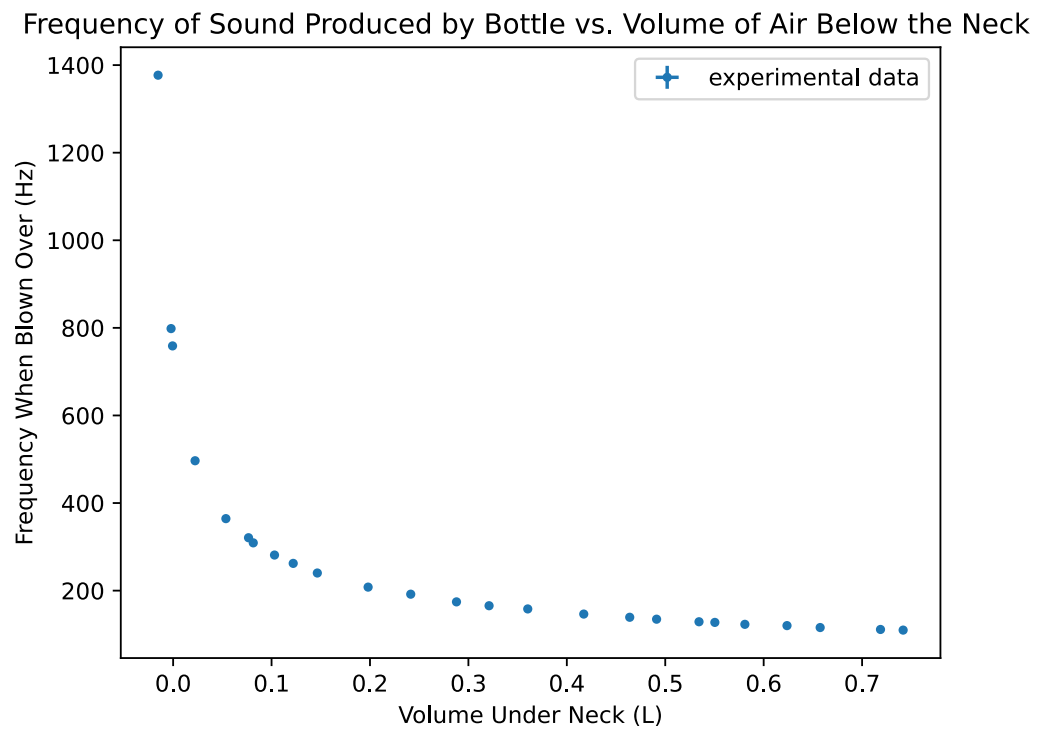


Figure 4: Plot of frequency produced by the bottle when blowing over the neck vs. the volume in the bottle under the neck (in the body). Note that error bars are included (as shown by the bars on the legend), however the error is too small to be shown outside the size of the dots. This is true for all of the following graphs.

- a) Add a model curve to your plot based on the equation you found in Q5 of Part I and the parameters for your bottle stated in Q1 of Part II. Note: this is **not** a fit, if the model does not line up perfectly with the data - that's ok; move to Q4. Include your plot in the space below.

Frequency of Sound Produced by Bottle vs. Volume of Air Below the Neck

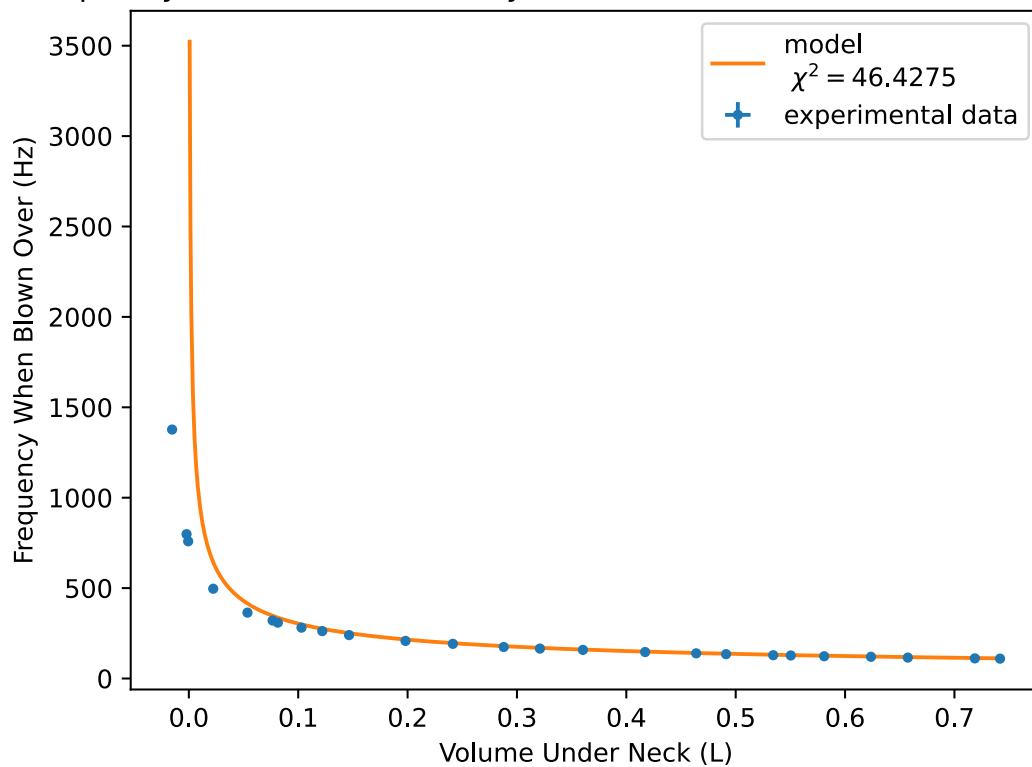


Figure 5: Plot of frequency produced by the bottle when blowing over the neck vs. the volume in the bottle under the neck (in the body) with a line representing the model.

4. [3 pts] Discuss the agreement between your data and model. Pay particular attention to anywhere the model starts to deviate from the data. Take up to 100 words max to explain why. Be concise and precise in your choice of words.

The model agrees with the data until the the sloped taper of the bottle towards the neck.

Frequency of Sound Produced by Bottle vs. Volume of Air Below the Neck

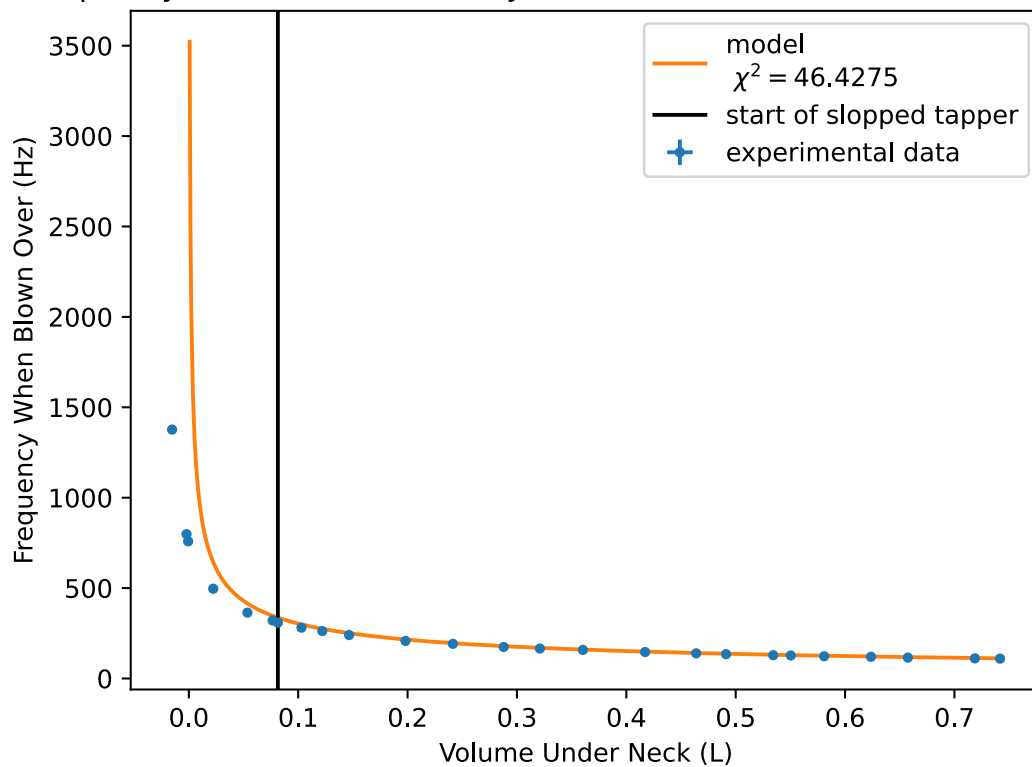


Figure 6: Plot of frequency produced vs. the volume in the bottle under the neck. The volume where the water reaches the slopped tapper is highlighted.

I believe this is due to underestimating of the volume of air acting as a spring, as reducing the volume results in a near-perfect fit.

Frequency of Sound Produced by Bottle vs. Volume of Air Below the Neck

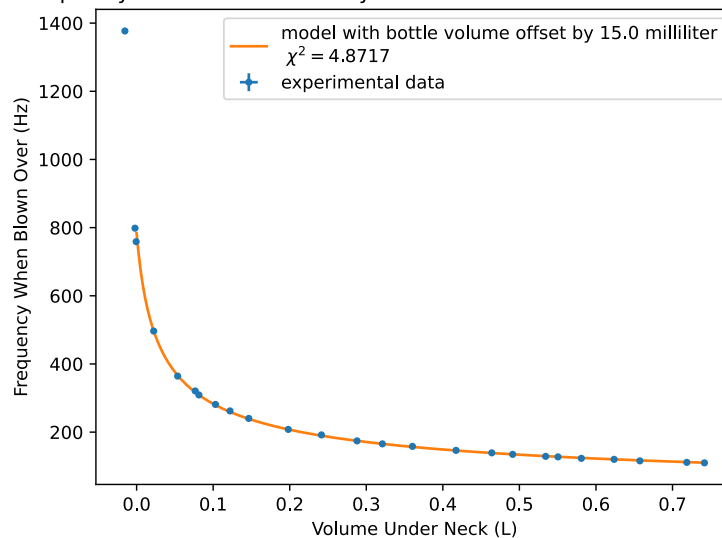


Figure 7: Plot of frequency produced vs. the volume in the bottle under the neck. The volume of air under the neck is reduced by 15 ml before the model is applied.

However, reducing the length of the neck to account for the offset results in a lower chi-squared, implying that some of the air is acting both as a mass and a spring.

This makes sense, as there is no hard boundary, and at 0 body volume, air must be acting in both ways for sound to be produced.

Words: 100

Bibliography

- [1] E. Edge, "Speed of Sound in Air and Other Gases." [Online]. Available: https://www.engineersedge.com/physics/speed_of_sound_13241.htm
- [2] E. Edge, "Properties of Water at Atmospheric Pressure Given in Imperial (I.P.) Metric SI Units." [Online]. Available: https://www.engineersedge.com/fluid_flow/properties_of_water_at_atmospheric_pressure_15859.htm